University of Ottawa

Department of Mathematics and Statistics

MAT 2143: Algebraic Structures
Professor: Hadi Salmasian

Midterm Exam

February 26, 2016

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(f) You are strongly recommended to write in **pen**, not pencil.
(g) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	7	Total
Maximum	5	3	3	3	5	5	3	27
Grade								

1. (a) **[2 points]** Write down the definition of a prime number.

Solution: An integer p is called prime if it satisfies the following conditions:

- $p > 1$
- If $d|p$ and $d > 0$ then $d = 1$ or $d = p$.

(b) **[3 points]** Prove that $\gcd(k^2 + 1, k^2 - 2) = 1$ for every integer k.

Solution: Suppose that $d > 0$ and $d|k^2 + 1$ and $d|k^2 - 2$. Then

$$d|(k^2 + 1) - (k^2 - 2) = 3 \;\Rightarrow\; d = 1 \text{ or } d = 3.$$

Next we show that $d = 3$ is impossible. To this end, we show that 3 does not divide $k^2 + 1$. Note that if $\overline{k}$ denotes the residue class of k in $\mathbb{Z}_3$, then

$$\overline{k^2 + 1} = \overline{k}^2 + \overline{1} \qquad \text{in } \mathbb{Z}_3.$$

Now we have the following table:

$\overline{k}$	$\overline{0}$	$\overline{1}$	$\overline{2}$
$\overline{k}^2 + \overline{1}$	$\overline{1}$	$\overline{2}$	$\overline{2}$

From the table it follows that for every $k \in \mathbb{Z}$ we have $\overline{k}^2 + \overline{1} \neq \overline{0}$ in $\mathbb{Z}_3$. This completes the argument.

2. **[3 points]** Prove the following theorem:

Let p be a prime number and let $m, n \in \mathbb{Z}$. If $p|mn$, then $p|m$ or $p|n$.

Solution: Write $d = \gcd(m, p)$. Then $d|p$, so $d = 1$ or $d = p$. If $d = p$ then $p|m$ and we are done. And if $d = 1$, then we can choose $x, y \in \mathbb{Z}$ such that $mx + py = 1$ (this is Bezout's identity). Now

$$mx + py = 1 \Rightarrow mnx + pny = n.$$

The left hand side of the last equality is divisible by p because $p|mn$. It follows that the right hand side is also divisible by p, hence $p|n$.

3. **[3 points]** Recall that

$$n! = n \times (n-1) \times \cdots \times 1$$

for every integer $n \geq 1$. Prove that every prime divisor of $2^{n!} + 1$ is strictly bigger than $n + 1$.

Hint: Use Fermat's Theorem.

Solution: Assume that $p \leq n + 1$ is a prime number such that $p | 2^{n!} + 1$. First note that $2^{n!} + 1$ is odd, therefore $p > 2$. It follows that $\gcd(p, 2) = 1$, so that by Fermat's theorem we have

$$2^{p-1} \equiv 1 \pmod{p}.$$

Since $p - 1 \leq n$, we obtain

$$2^{n!} \equiv (2^{p-1})^{\frac{n!}{p-1}} \equiv 1^{\left(\frac{n!}{p-1}\right)} \equiv 1 \pmod{p}.$$

It follows that $2^{n!} + 1 \equiv 2 \not\equiv 0 \pmod{p}$ since $p > 2$, which is a contradiction.

4. **[3 points]** Determine all $x \in \mathbb{Z}$ such that

$$x \equiv -1 \pmod 7 \quad \text{and} \quad x \equiv 1 \pmod{15}.$$

You should justify your answer.

Solution: By the method that is based on the Chinese Remainder Theorem, first we need to find $p, q \in \mathbb{Z}$ such that

$$7p + 15q = 1.$$

To this end, we use the extended Euclidean algorithm (or direct guessing):

r_i	15	7	1
s_i	1	0	1
t_i	0	1	-2

which results in $p = -2$ and $q = 1$. Now the Chinese Remainder Theorem implies that the simultaneous solutions are of the form

$$x \equiv 7(-2)(1) + 15(1)(-1) \equiv -29 \pmod{105}.$$

5. (a) **[2 points]** Write down the definition of a monoid.

(b) **[1 point]** Give an example of a monoid which is not a group. You should justify your answer.
(c) **[2 points]** Prove the following theorem:

If M is a monoid, then the set of all invertible elements of M is a group (with the binary operation induced from M).

Solution: (a) A monoid is a set with a binary operation which is associative and has an identity element.

(b) For example, $M := \mathbb{N} \cup \{0\}$ with addition is a monoid with identity element $0 \in M$, but positive elements of M are not invertible, hence M is not a group.

(c) Denote the defined subset by M^*. Let $e \in M$ be the identity element of M. Then $e \in M^*$ because $e \star e = e$, hence e is invertible in M. Next we show that M^* is closed under $\star$. Choose $a, b \in M^*$. Then

$$\exists_{a',b' \in M} \ : \ a \star a' = a' \star a = e \text{ and } b \star b' = b' \star b = e.$$

Now

$$(a \star b) \star (b' \star a') = a \star (b \star b') \star a' = a \star e \star a' = a \star a' = e.$$

Similarly, $(b' \star a') \star (a \star b) = e$, hence $a \star b$ is invertible, and consequently, $a \star b \in M^*$.

Associativity of M^* is inherited from M. Finally, we show that every $a \in M^*$ is invertible in M^*: If an element $a \in M$ is invertible, then

$$\exists_{a' \in M} \ : \ a \star a' = a' \star a = e.$$

It follows immediately by definition that a' is invertible with inverse a. Thus $a' \in M^*$. Hence a is invertible in M^*.

6. Let $\sigma \in S_9$ denote the permutation given by

$$\sigma = (3\ 1\ 7)(9\ 8\ 1\ 4)(3\ 1\ 5).$$

(a) **[2 points]** Write down the decomposition of σ as a product of disjoint cycles.

(b) **[1 point]** Determine the smallest integer $d > 0$ such that $\sigma^d = \varepsilon$. You should justify your answer.

(c) **[2 points]** Is it possible to write σ as a product of (not necessarily disjoint) 5-cycles? You should justify your answer.

Solution:

(a) The matrix representation of σ is

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 2 & 4 & 9 & 1 & 6 & 3 & 7 & 8 \end{pmatrix}$$

Thus

$$\sigma = (1\ 5)(3\ 4\ 9\ 8\ 7).$$

(b) For the transposition $\tau + (1\ 5)$ we have $\tau \neq \varepsilon$ but $\tau^2 = \varepsilon$. For the 5-cycle $\eta = (3\ 4\ 9\ 8\ 7)$ we have $\eta^5 = \varepsilon$ but $\eta^\ell \neq \varepsilon$ for any $0 < \ell < 5$. It follows that $\tau^n = \varepsilon$ exactly for all integers n such that $2|n$, and $\eta^n = \varepsilon$ exactly for all integers n such that $5|n$. Since τ and η are disjoint, the smallest $d > 0$ such that $\sigma^d = \varepsilon$ is $d = 10$.

(c) Note that σ is an odd permutation (this follows immediately from the definition of σ, because 3-cycles are odd and 4-cycles are even). However, a 5-cycle is always even, and a product of even permutations is always an even permutation (this is easy by the definition of even and odd permutations). Therefore it is impossible to write σ as a product of 5-cycles.

7. Let G be a group and let $e \in G$ denote the identity element of G. Suppose that $x^2 = e$ for every $x \in G$.

(a) **[2 points]** Prove that G is abelian.

(b) **[1 point]** Give an example of such a group G other than the trivial group or the group $\mathbb{Z}_2$.

Solution: For every $x, y \in G$ we have:

$$(xy)^2 = \varepsilon \;\Rightarrow\; x(xyxy)x = x\varepsilon y = xy \;\Rightarrow\; x^2(yx)y^2 = xy \;\Rightarrow\; yx = xy.$$

This proves that G is abelian.

(b) **[1 point]** For instance $G \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.